

Frequently Bought Together

...and the three things it can't do

60 minutes · no prerequisites · you will not be asked to write any code

AI Engineering · business engagement

Would you have shipped this?

A real recommendation from the live tool

VINTAGE UNION JACK BUNTING

recommends →

A SEASONAL LINE THAT WAS IN EVERY BASKET

Confidence 0.61 — looks convincing

Lift 1.02 — the pairing is no more likely than chance

Hold that thought

Every number on this slide is real.

By 0:25 you will be able to say in one sentence why this row is worthless.

Confidence is the number people read. Lift is the number that decides.

Three reasons a recommendation goes wrong

In order of how often we see them

1

Too few baskets

A pairing built on 30 baskets can be a coincidence. The basket count is your sample size - read it first, every time.

2

A promotion

If two products were in the same offer, they look associated for as long as that period sits in the data.

3

It isn't behaviour at all

Some rows come from the product description text, not from anything anyone bought. They are labelled.

None of these are the model being clever or stupid. They are all the data telling the truth about something you didn't ask about.

Reading a recommendation

One real row, one number at a time

CHILDS GARDEN SPADE PINK → CHILDS GARDEN SPADE BLUE

40

baskets

Your sample size. Read this first — a huge lift on 8 baskets is noise.

0.85

confidence

Of baskets with the pink spade, 85% also had the blue one.

234

lift

234× more likely than chance. This is the ranking column.

Now go back to the bad row: confidence 0.61, lift 1.02. Lift near 1 means "chance". That's the whole answer.

How to read lift

The only column you sort by

~ 1

No relationship

They co-occur about as often as chance predicts. Ignore.

2 – 5

Mild association

Real but weak. Fine for a "you might also like" slot.

10 +

Strong association

Usually a genuine complement.

100 +

Variant or set

Colour/size variants, or two halves of one purchase.

Why not sort by confidence? A product that's in a third of all baskets shows high confidence against everything. Lift corrects for that.

Two signals feed the list — and they are not equal

Check the method column before you build a plan on a row



co_purchase

Built from what people actually bought together.

This is the real signal. Trust it in proportion to the basket count.

Covers about 30% of the catalogue.



content_tfidf

Built from the product description text.

Nobody has been observed buying these together.
They simply read alike.

A starting point — never evidence.

Why keep the weaker signal at all? Without it, 7 products in 10 show an empty panel — and people generalise from the first empty screen they see.

Your turn — 15 minutes, in pairs

Trust it / don't trust it / need more information. Five real rows.

#	Pair	Baskets	Confidence	Lift
1	Childs Garden Spade Pink → Blue	40	0.85	234
2	Landmark Frame Covent Gdn → Oxford St	30	0.77	323
3	A new line, method = content_tfidf	—	—	—
4	Silk Fan → Lunch Bag Vintage Doily	34	0.12	2.4
5	Seasonal best-seller → almost everything	410	0.44	1.1

Three things it cannot do

Said first, and unprompted — the credibility of everything else depends on it

1 It cannot tell you why

Bunting and paper plates co-occur because both are party purchases — not because one drives the other. Placing them together is sensible. Concluding that discounting one lifts the other is not.

2 It cannot see what you didn't stock

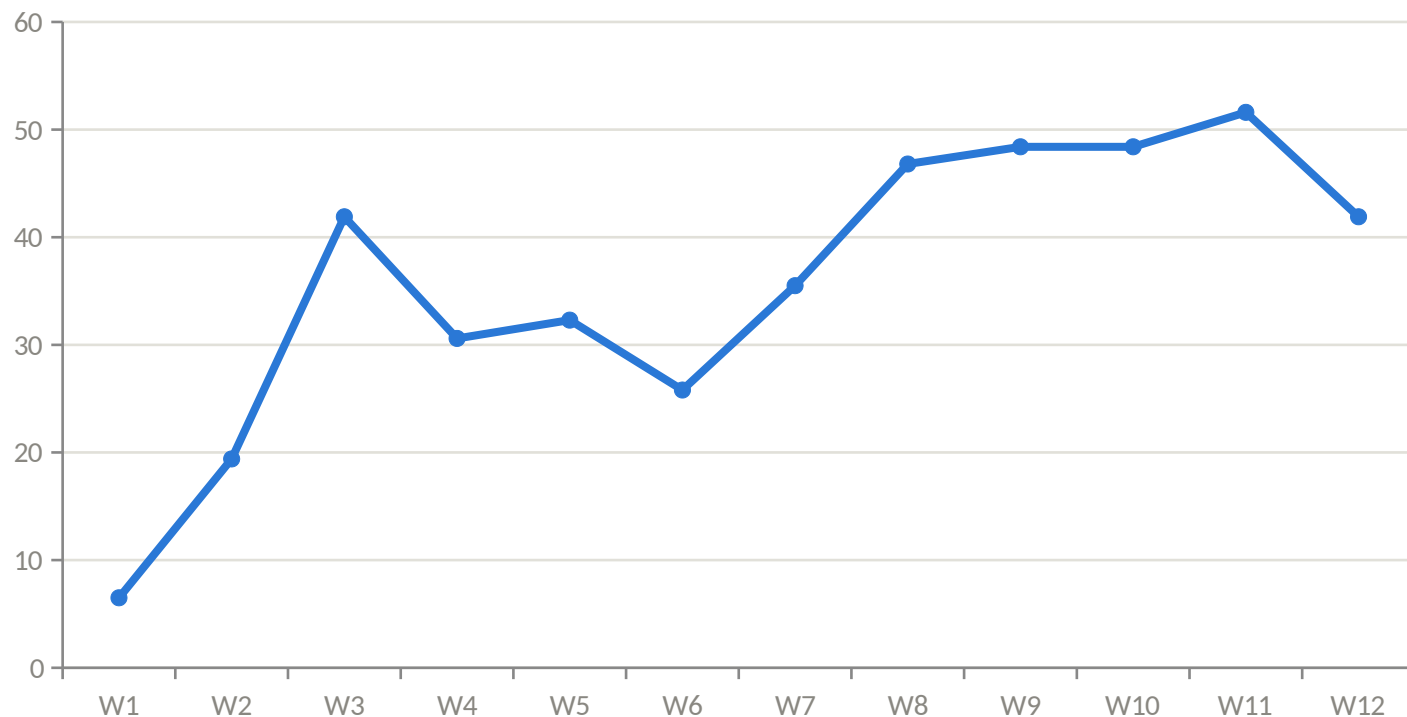
A product out of stock for six weeks builds its pairings from the weeks it was available. It will look less connected than it is.

3 It cannot decide anything

Nothing writes to a trading system. Every recommendation is a suggestion a person accepts or rejects. That is a design decision, not a limitation.

How it's going

Weekly reach — % of the 62 licensed users active that week



The two jumps

Week 3 and week 8 are these workshops.

19% → 42% after the first.

36% → 47% after the second.

Both held rather than decaying — which is the part that matters. A spike that decays means people looked once.

Week 6 is the school holidays. We report the dips too — a chart with no dips is a chart nobody believes.

One ask

Use it once this week on a real decision — and tell us what happened.

Feedback button

Two clicks and a score. We read every one — the June filtering change came from this.

Something looks wrong?

Say so, and include both product codes. More useful than a low score with no comment.

Want a session for your team?

Reply to the monthly update. Booked within a week.